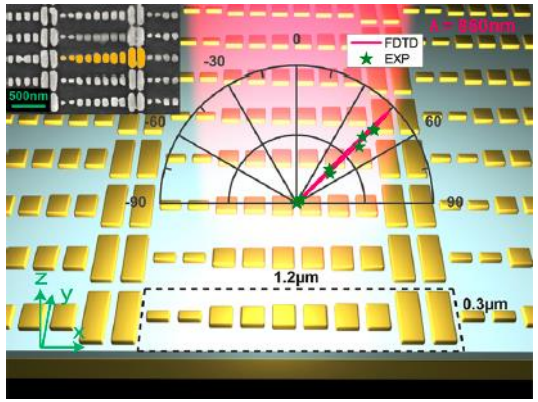
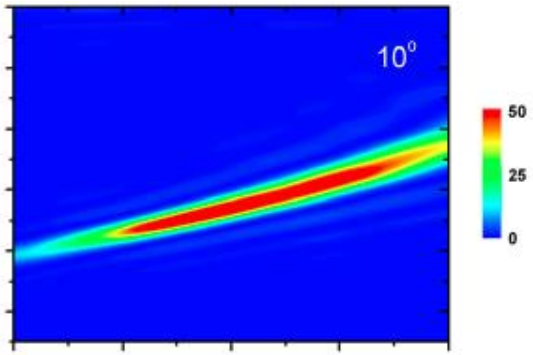
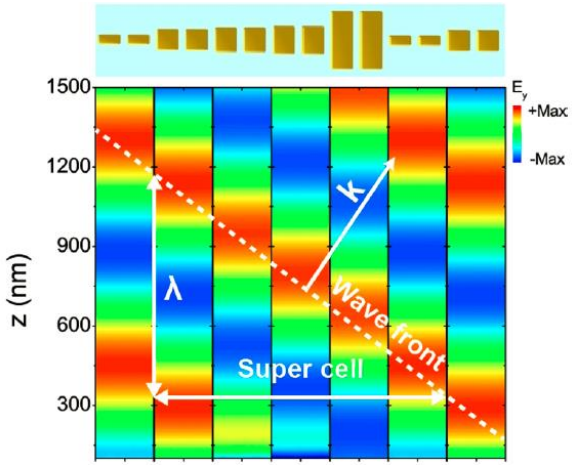
http://dx.doi.org/10.1021/nl3032668

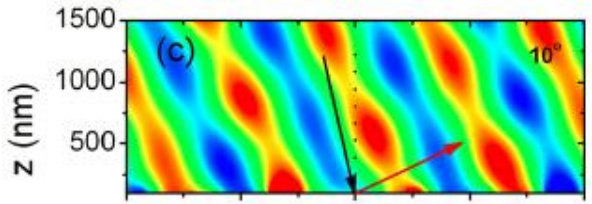
异常反射

本节课涉及的论文发表在 *Nano Letters* 上









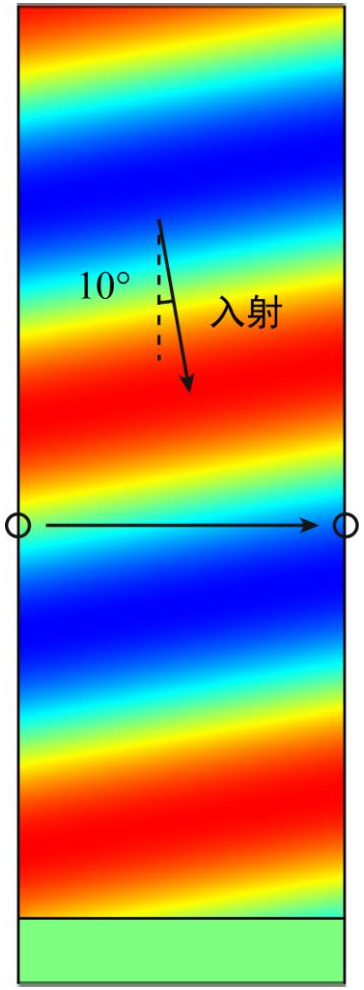
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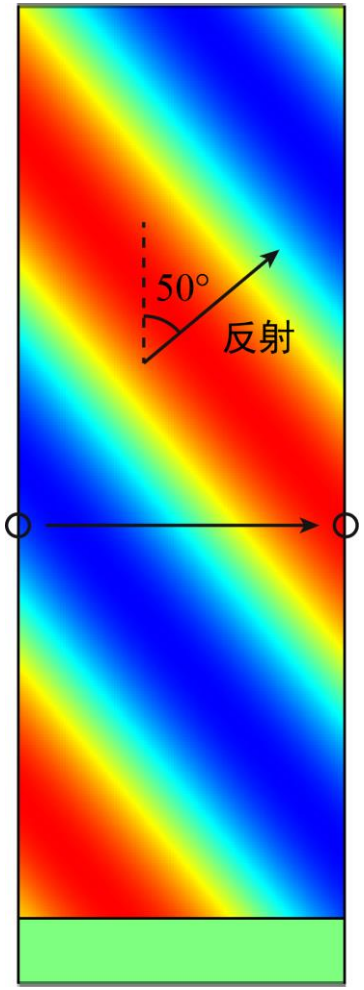
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满足单波长、斜入射

能用 Bloch 边界条件吗？

周期性条件、Bloch条件、BFAST光源等知识已经在专题三中详细介绍。





由于不满足正常的反射定律，所以左右两个边界上对于入射光和反射光的相位差是不同的，所以不能用Bloch边界条件

这也正是“反常”之所在

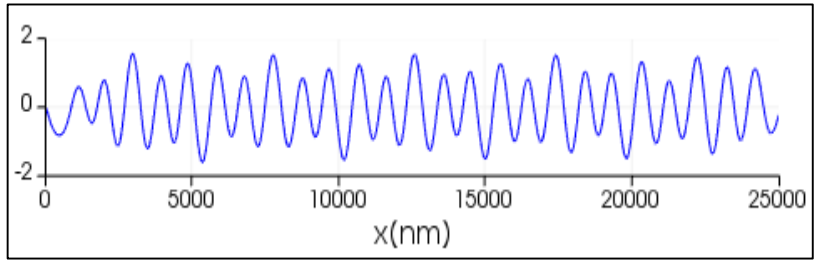
时间域 → 频率域的一维傅里叶变换：

$$F(\omega) = \int f(t)\mathrm{e}^{-\mathrm{i}\omega t}\mathrm{d}t$$

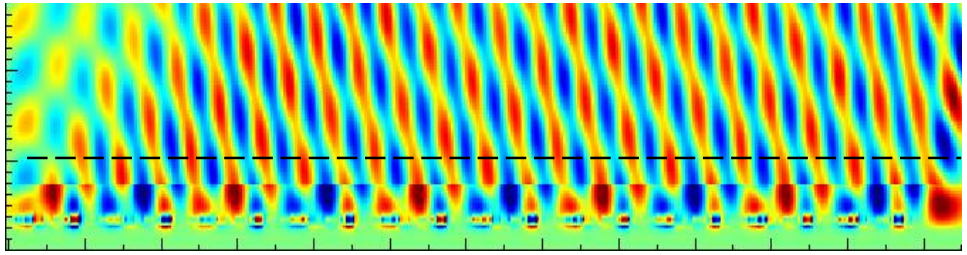
坐标空间 → 波矢空间的一维傅里叶变换：

$$F(\frac{2\pi}{\lambda}) = F(k) = \int f(x)\mathrm{e}^{-\mathrm{i}kx}\mathrm{d}x$$

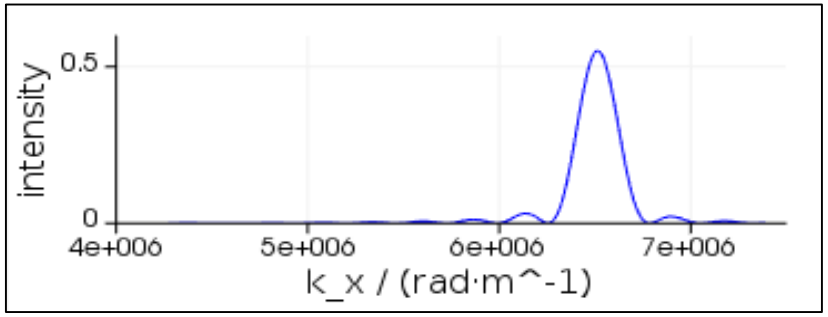
其中 $k = 2\pi/\lambda$ 为波矢



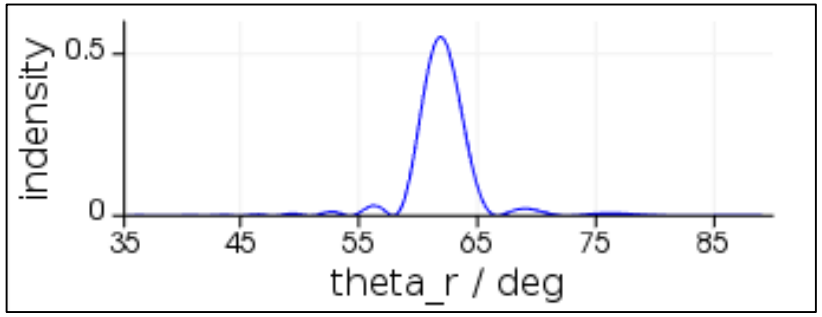
黑色虚线处的 E_y



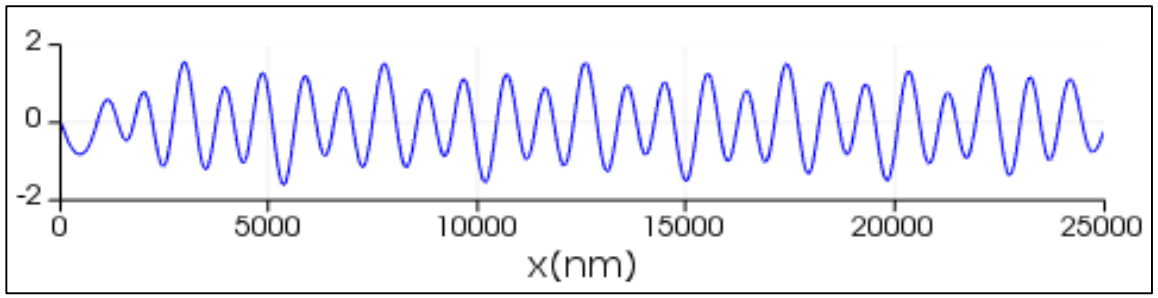
1D 傅里叶变换
得到 x 方向上的波矢



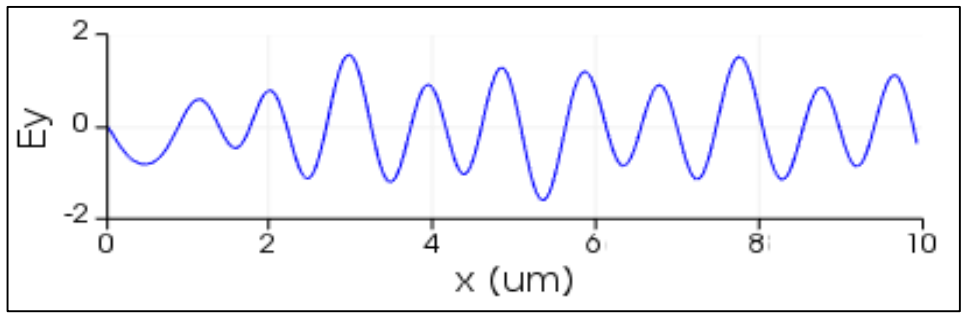
$$k_x = k_0 \sin(\theta_r)$$



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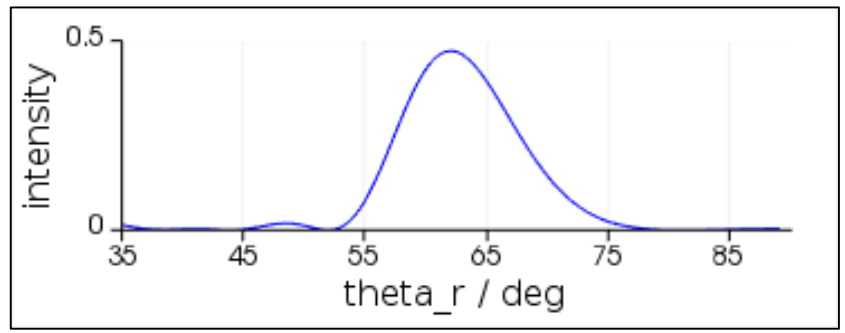


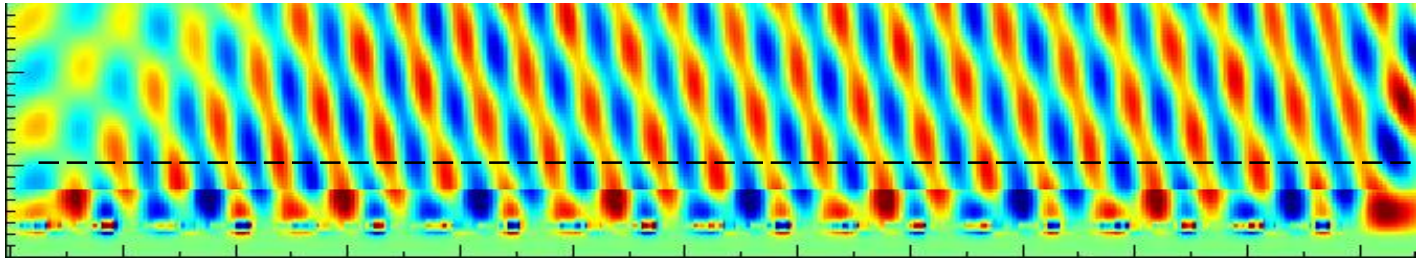
如果数据量减少
例如只画8个超胞



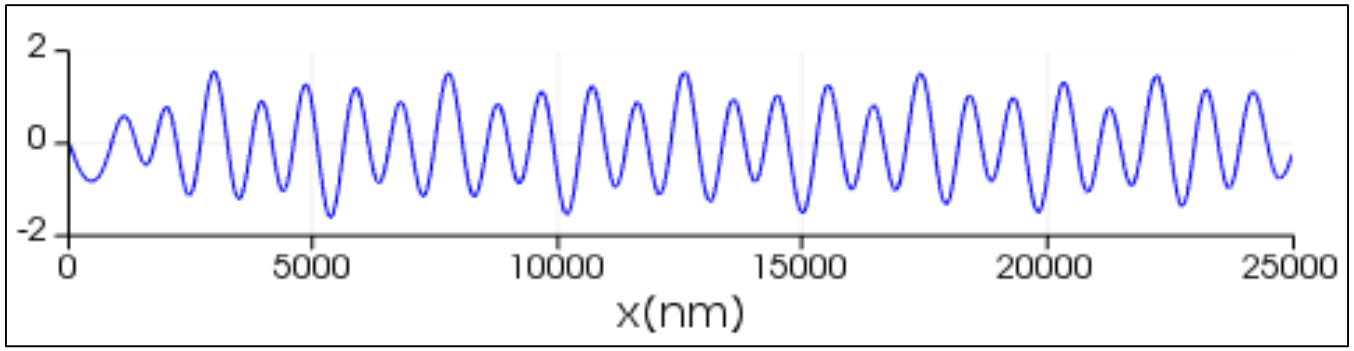
1D 傅里叶变换，然后利用

公式： $k_x = k_0 \sin(\theta_r)$





黑色虚线处的 E_y

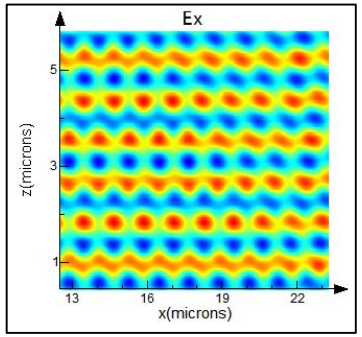


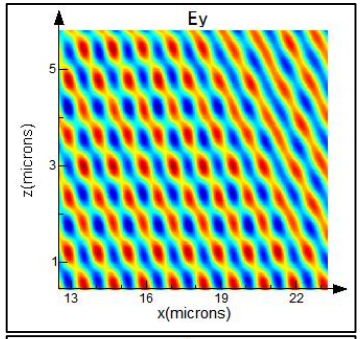
单凭黑色虚线处的电场是不能判断光是斜向右上

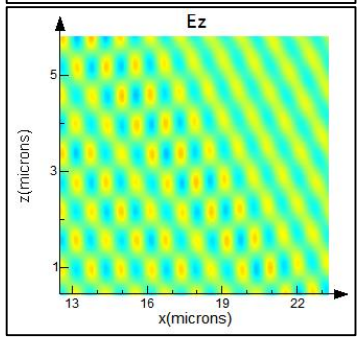
还是斜向左上传播的

因为1D傅里叶变换丢失了 z 方向的波动信息

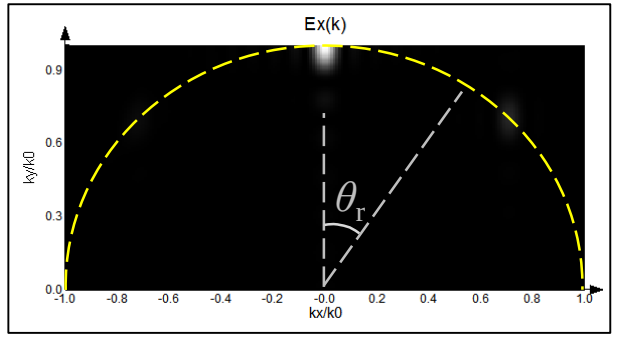
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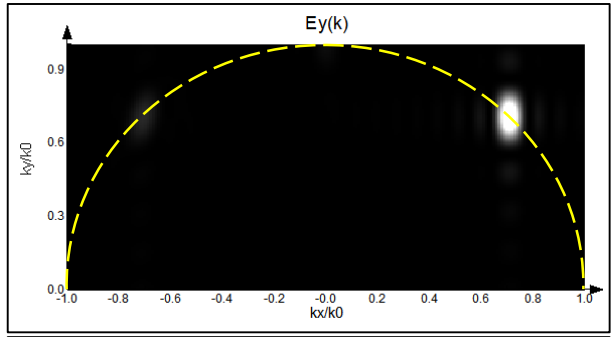


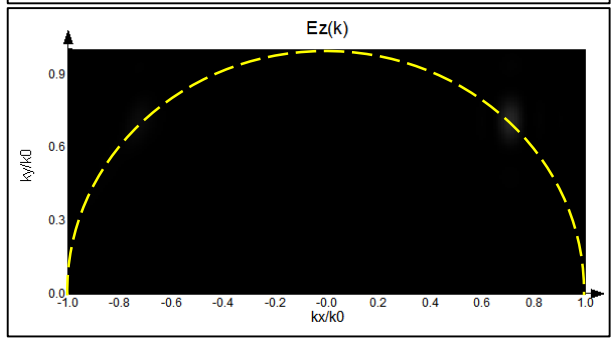




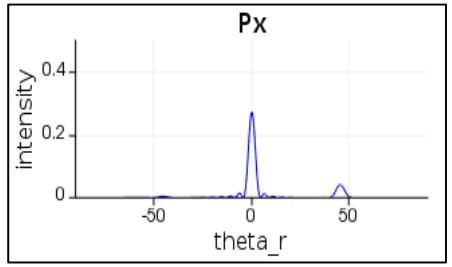
二维
傅里叶变换

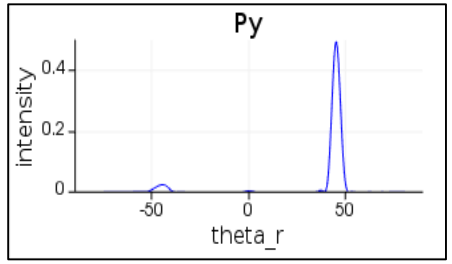


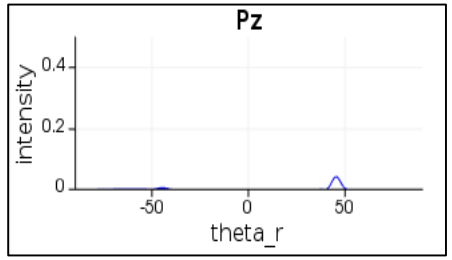




黄色虚
线处的值







叠加

